# THE HOMOTOPY COLIMIT OF A SUBSEQUENCE MAPS ISOMORPHICALLY TO THE HOMOTOPY COLIMIT

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 10448-10514, following the statement at 10430-10446.

*Editorial note.* The h identities are 'the reader verifies' (10507): mechanical index bookkeeping from the displayed component formulas; the square-zero step is a '(small argument omitted)' instantiation of lemma-third-map-square-zero, recorded.

*Editorial note.* Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 10510 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

*Allowed standard prerequisites (editorial).*

- Homotopic maps agree after localization. lemma-third-isomorphism-triangle, cited at proof line 10496; complete proof at source lines 641-670, queued for separate carding. Supplies that c a and d b being identities makes the induced map on homotopy colimits an isomorphism one way.
- Nilpotent perturbations of the identity. lemma-third-map-square-zero, cited at proof line 10510; complete proof at source lines 710-728, queued for separate carding. Supplies that id - psi phi having square zero forces psi phi (hence phi, psi) to be isomorphisms.

*Source reference.* Original lemma 014A: sources/stacks-derived/pinned-original-source.tex, source lines 631–631. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 05QP: sources/stacks-derived/pinned-original-source.tex, source lines 701–701. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $\mathcal{D}$ be a triangulated category. Let $(K_n, f_n)$ be a system of objects of $\mathcal{D}$. Let $n_1 < n_2 < n_3 < \ldots$ be a sequence of integers. Assume $\bigoplus K_n$ and $\bigoplus K_{n_i}$*

*Date*: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*exist. Then there exists an isomorphism* $hocolim K_{n_i} \to hocolim K_n$ *such that*

$$\begin{array}{ccc} K_{n_i} & \longrightarrow & hocolim K_{n_i} \\ {\scriptstyle id}\downarrow & & \downarrow \\ K_{n_i} & \longrightarrow & hocolim K_n \end{array}$$

*commutes for all* $i$.

*Proof.* Let $g_i : K_{n_i} \to K_{n_{i+1}}$ be the composition $f_{n_{i+1}-1} \circ \ldots \circ f_{n_i}$. We construct commutative diagrams

$$\begin{array}{ccc} \bigoplus_i K_{n_i} & \xrightarrow{1-g_i} & \bigoplus_i K_{n_i} \\ {\scriptstyle b}\downarrow & & \downarrow{\scriptstyle a} \\ \bigoplus_n K_n & \xrightarrow{1-f_n} & \bigoplus_n K_n \end{array} \quad \text{and} \quad \begin{array}{ccc} \bigoplus_n K_n & \xrightarrow{1-f_n} & \bigoplus_n K_n \\ {\scriptstyle d}\downarrow & & \downarrow{\scriptstyle c} \\ \bigoplus_i K_{n_i} & \xrightarrow{1-g_i} & \bigoplus_i K_{n_i} \end{array}$$

as follows. Let $a_i = a|_{K_{n_i}}$ be the inclusion of $K_{n_i}$ into the direct sum. In other words, $a$ is the natural inclusion. Let $b_i = b|_{K_{n_i}}$ be the map

$$K_{n_i} \xrightarrow{1,\ f_{n_i},\ f_{n_i+1} \circ f_{n_i},\ \ldots,\ f_{n_{i+1}-2} \circ \ldots \circ f_{n_i}} K_{n_i} \oplus K_{n_i+1} \oplus \ldots \oplus K_{n_{i+1}-1}$$

If $n_{i-1} < j \leq n_i$, then we let $c_j = c|_{K_j}$ be the map

$$K_j \xrightarrow{f_{n_i-1} \circ \ldots \circ f_j} K_{n_i}$$

We let $d_j = d|_{K_j}$ be zero if $j \neq n_i$ for any $i$ and we let $d_{n_i}$ be the natural inclusion of $K_{n_i}$ into the direct sum. In other words, $d$ is the natural projection. By TR3 these diagrams define morphisms

$$\varphi : \text{hocolim} K_{n_i} \to \text{hocolim} K_n \quad \text{and} \quad \psi : \text{hocolim} K_n \to \text{hocolim} K_{n_i}$$

Since $c \circ a$ and $d \circ b$ are the identity maps we see that $\varphi \circ \psi$ is an isomorphism by Lemma 014A. The other way around we get the morphisms $a \circ c$ and $b \circ d$. Consider the morphism $h = (h_j) : \bigoplus K_n \to \bigoplus K_n$ given by the rule: for $n_{i-1} < j < n_i$ we set

$$h_j : K_j \xrightarrow{1,\ f_j,\ f_{j+1} \circ f_j,\ \ldots,\ f_{n_i-1} \circ \ldots \circ f_j} K_j \oplus \ldots \oplus K_{n_i}$$

Then the reader verifies that $(1 - f) \circ h = \text{id} - a \circ c$ and $h \circ (1 - f) = \text{id} - b \circ d$. This means that $\text{id} - \psi \circ \varphi$ has square zero by Lemma 05QP (small argument omitted). In other words, $\psi \circ \varphi$ differs from the identity by a nilpotent endomorphism, hence is an isomorphism. Thus $\varphi$ and $\psi$ are isomorphisms as desired. $\square$